

Examination of stand structure on quail plantations in the Red Hills region of Georgia and Florida managed by the Stoddard–Neel system: an example for forest managers

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Summary

Longleaf pine (*Pinus palustris* Mill.) on quail hunting plantations in the Red Hills region of southern Georgia and northern Florida, USA, has been managed over the last 40 years by the ‘Stoddard–Neel’ system. Developed during World War II, the system focuses on stand structures with low densities and an open midstorey. These structures benefit both hunting and ecological concerns such as herbaceous diversity. We documented the forest stand structure on quail plantations in the Red Hills and compared it to the traditional ‘reverse-J’ structure. We conducted a variable plot cruise of the study area at an intensity of one plot per 4 hectares. Using these preliminary cruise data, we selected 22 sets of 40-ha measurement blocks. On the 40-ha blocks, we measured tree size and location in fixed-area plots on three randomly chosen transects in each block. The number of trees decreases rapidly as the diameter increases, up to a point. Next, there is a ‘bump’ or second mode in the larger diameter classes, after which the number of trees again decreases. Instead of the classic ‘reverse-J’ diameter distribution of shade-tolerant, uneven-aged forests, this distribution is bimodal. This forest-wide structure was observed at the plantation and block level, with observed variation greatly increasing at the transect and plot levels. The shape of the diameter distributions reflects a preferential removal of saplings and midstorey trees for understorey diversity, aesthetics and quail management. These data further suggest that uneven-aged forests can be managed at the smaller scales as two-aged (‘two-sized’) stands. This method also has the advantage of concentrating growing space on larger, higher-value trees.

Introduction

When William Bartram roamed the southeastern United States in the eighteenth century, the coastal plain uplands were covered with large, variably stocked longleaf pine (*Pinus palustris* Mill.) stands (Bartram, 1791). Longleaf pine was the predominant species spread over 24–36 million ha before European settlement (Frost, 1993). Large-scale (e.g. hurricanes) and medium-scale (e.g. tornadoes) catastrophic events caused many areas to develop as even-aged stands. Smaller scale disturbances, e.g. lightning strikes and fires of varying intensity, caused stands to develop which contained a variety of age and diameter classes (Schwarz, 1907; Wahlenberg, 1946; Platt *et al.*, 1988; Landers *et al.*, 1995). Pre-European-settlement fires have been estimated to occur every 1–10 years (Chapman, 1932; Christensen, 2000). Longleaf pine is well-suited to multi-aged forest conditions because it tolerates fire at all age classes better than its pine and hardwood competitors can (Boyer, 1990), although fire will always impact growth (Boyer, 2000), and because it can respond to release at almost any age (Boyer, 1990).

The role of uneven-aged management and the value of continuously standing forest cover in southern pine ecosystems has been the subject of considerable discussion (Platt *et al.*, 1988). Although the value to ecosystem health and biodiversity of standing pine forests has largely been accepted (Simberloff, 1993; Christensen, 2000; Platt *et al.*, 1988), the methodology for attaining and maintaining these structures is open to debate (Rudolph and Conner, 1996; Engstrom *et al.*, 1996).

Attempts at quantifying multi-aged forests have assumed that a balanced forest has a reverse-J shaped diameter distribution (Meyer, 1943, 1952) that can be described by a ratio of succeeding diameter classes (called a 'q-ratio') (de Liocourt, 1898). While some may argue that the shape of the curve refers to the diameter distribution of forests, not stands, in practice foresters have sought to implement this structure at the smallest scale possible, even in shade-intolerant species (Baker *et al.*, 1996; Farrar, 1996).

However, trying to maintain such a structure can devalue the very ecological attributes that make longleaf pine ecosystems important. In

these forests, the diversity lies in the understorey. In one 'old-growth' type forest, the Wade Tract Preserve in Thomas County, Georgia, species diversity of up to 40 species per square metre has been documented (Sharon Hermann, Tall Timbers, personal communication). Overall diversity is reduced by competition from a dense overstorey and midstorey, a necessary result of a constant q-factor in a reverse-J structure. Large old pines are considered valuable both for their influence on the rest of the ecosystem and as habitat for species of special concern. Trees are moved through the midstorey in patches as opposed to trying to maintain a widespread midstorey.

A unique alternative for creating and managing uneven-aged stands is practised in the Red Hills region of Florida and Georgia, situated between Tallahassee, Florida and Thomasville, Georgia. Dubbed the 'Stoddard–Neel System' after its practitioners Herbert Stoddard and Leon Neel, it professes to use single-tree selection to create a multi-aged forest that provides an open midstorey and a great deal of growing space to the understorey, thereby maintaining suitable habitat for species of interest, such as northern bobwhite quail (*Colinus virginianus*), gopher tortoise (*Gopherus polyphemus*) and red-cockaded woodpecker (*Picoides borealis*).

This study examines two hypotheses. The first is that the Stoddard–Neel structure significantly differs from the standard 'reverse-J' diameter structure. The second hypothesis is that the Stoddard–Neel structure is present at all scales of management, from the stand level to the landscape level.

History of quail plantation forest management in the Red Hills

In the Red Hills region of southern Georgia and nearby Florida, typical pristine areas are dominated by longleaf pine and wiregrass (*Aristida beyrichiana*). Disturbed areas are dominated by loblolly (*Pinus taeda*) and shortleaf (*P. echinata*) pines and old-field grasses. Various hardwoods and woody shrubs are present in densities dependent upon the local soil moisture and prescribed fire regime (Thomas College, 1994).

Individuals seeking land on which to grow row

crops and graze cattle settled on cutover timberland in the Red Hills of south Georgia and North Florida in the early nineteenth century. After many of the properties were first fragmented into tenant farms, they were then aggregated into quail plantations around the beginning of the twentieth century. Timber harvesting was limited during the first half of the twentieth century. Additionally, in the context of maintaining healthy quail habitat, basal areas were largely kept below $15 \text{ m}^2 \text{ ha}^{-1}$ to ensure enough growing space for the understorey plants that provide food for the bobwhite (Moser and Palmer, 1997).

In the 1920s, a biologist, Herbert Stoddard, documented the life history of quail and discovered the importance of burning in maintaining appropriate quail habitats (Stoddard, 1931). During World War II, Stoddard devised the 'Stoddard System' of selection system management to supply wood for the war effort while still maintaining a forest overstorey. In 1950, Leon Neel started working with Stoddard and continued his consulting business after Stoddard retired.

Methods

In order to determine the variability of basal area, we conducted a variable plot cruise using a 10 BAF (English units) prism, of five plantations in the study area at an intensity of one prism point per 4 ha. Plots were spaced 200 m apart and marked for future use. These plots stayed largely on the upland sites. Using these preliminary cruise data, 22 sets of 40-ha blocks were placed – nine in 1998 (Plantations BL and DK); seven in 1999 (Plantation SG); six in 2000 (Plantations MR and SW) – such that the mean basal area of each block fitted into one of three basal area categories: (a) low, $0\text{--}7 \text{ m}^2 \text{ ha}^{-1}$; (b) medium, $7\text{--}15 \text{ m}^2 \text{ ha}^{-1}$; and (c) high, $>15 \text{ m}^2 \text{ ha}^{-1}$. Two blocks, Block 1, Plantation MR and Block 2, Plantation SW, were smaller than 40 ha because they were bounded on four sides by other ownerships and/or swamps. Block 1, Plantation SW was ~ 40 ha, but was not square. Four of the properties were managed by Mr Neel and one (BL) was managed by another consultant in a very similar manner.

Each 40-ha block contained seven or eight possible transects ~ 100 m apart. In most of the blocks, we randomly selected three transects and

placed 0.04 ha plots (0.08 ha plots in 1999 and 2000) 100 m apart. Where the blocks were not square, we added transects to keep the number of plots per block approximately equal. In each of these plots, we measured tree size (height, diameter) and location.

We examined what levels of aggregation matched the forest-wide distribution. By comparing diameter distributions at the forest-wide (entire study), ownership ('plantation'), measurement block (40 ha), transect and plot levels, we hoped to determine the minimum management unit that foresters should aim for when practising the Stoddard–Neel system. Accordingly, we analysed the variation in the data by comparing the χ^2 distribution test at each level of aggregation (Zar, 1999) (Table 1). We took the forest-wide diameter distribution and converted it into proportions for each diameter class. The expected number of observations for a diameter class is equal to the size of the test data set multiplied by these proportions. In this case the forest-wide proportions represent the standard Stoddard–Neel distribution, which we compared against the plantations' diameter distributions; then the plantation was compared to its blocks, each block was compared to its transects and, finally, each transect was compared to its constituent plots.

Results

Most of the stands in the study had low to medium basal area. The blocks of high-density stands had the greatest mean tree height and the largest mean tree diameter, whereas the blocks of medium-density stands had the highest number of trees per hectare and the smallest average diameter. The forests were broken up by varying proportions of roads, fields and food plots, which are necessary for quail management.

Diameter distribution at plantation level versus the 'reverse-J' distribution

Under the Stoddard–Neel system, the number of trees decreases rapidly as the diameter increases, up to a point. Next there is a 'bump' or second mode in the larger diameter classes, after which the number of trees again decreases. This

Table 1: Chi-squared analysis of the diameter distributions of the entire study (the landscape level 'Stoddard-Neel' distribution) and different scales from five Red Hills plantations ('BL' through 'SW'), all 22 40-ha measurement blocks, 72 transects and 404 plots

	Landscape vs. Plantation	Plantation vs. Block	Block vs. Transect	Transect vs. Plot
Total comparisons	5	22	72	404
Comparison with significant differences	0 (0%)	0 (0%)	5 (7%)	161 (40%)

Note that none of the plantations were found to be significantly different from the landscape-level distribution, and none of the blocks were found to be significantly different from their parent plantations. A higher proportion of transects and even more plots show significant differences.

distribution is bimodal in character. The open midstorey is the result of fire and mechanical control of sapling ingrowth and a preference of the Stoddard-Neel system for retaining large trees.

Figure 1 also shows the line of a reverse-J shaped graph (at $q = 1.1$, with 1 cm diameter classes) with a basal area of $12 \text{ m}^2 \text{ ha}^{-1}$ and a

maximum diameter of 70 cm, one tree of this diameter being found on every 11 ha. The diameter distribution of the plantations differs from the distribution of the reverse-J curve. Given the quail-hunting constraints of a low residual basal area and an open midstorey, the Stoddard-Neel system is a more appropriate structure in these longleaf pine stands.

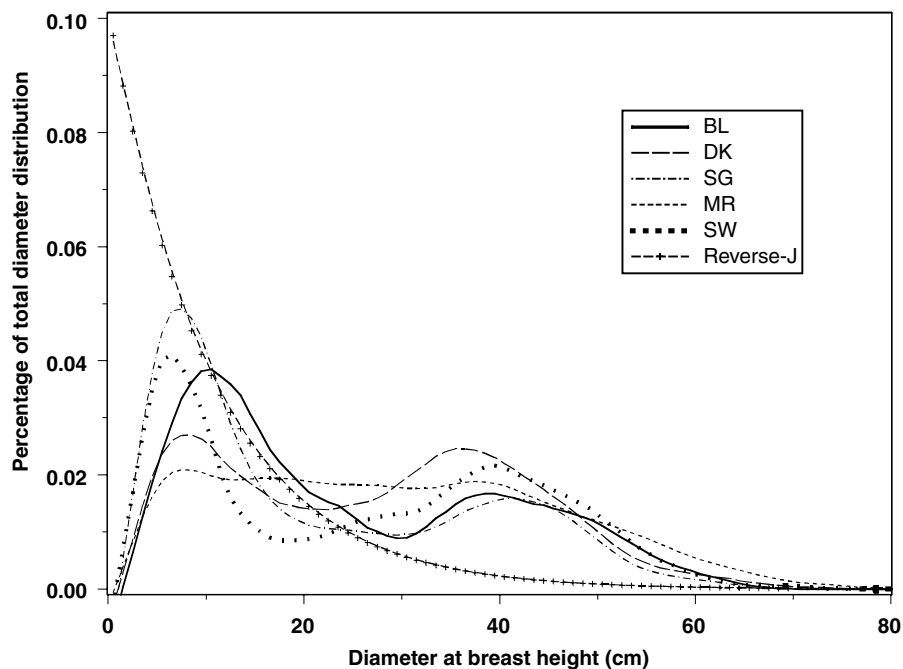


Figure 1. Diameter distribution percentages by plantation, showing the typical Stoddard-Neel structure. Curves were plotted using Friedman super-smoothing (Insightful Corporation, 2001). A theoretical reverse-J curve ($B = 12 \text{ m}^2 \text{ ha}^{-1}$, $D = 70 \text{ cm}$, $q = 1.1$, 1 cm diameter classes) is superimposed over the actual plantation-level values.

Diameter distribution across measurement scales

We then compared the relationship between size classes: forest-wide vs. plantation, plantation vs. block, block vs. transect and transect vs. plot. We compared only those scales that overlapped, e.g. block MR1 and transect MR1.3, or transect MR1.3 and plot 319 (a plot on that transect) (Table 1). None of the five plantations were significantly different from the forest-wide (aggregated) distribution and none of the 22 blocks were significantly different from their parent plantation distribution. Mean density of the overstorey in the blocks had no influence over whether the block diameter distribution was significantly different from the parent plantation or the subordinate transect. In fact, only two of the 22 blocks were significantly different from the forest-wide distribution (unpublished data). Five of the 72 transects (7 per cent) were significantly different from their parent blocks. Of the 404 plots compared to their parent transects, 161 were different, ~40 per cent. There is greater variation from the forest-wide distribution at the transect and plot levels than at the block level, and a more distinctly two-storeyed (bimodal) structure. At the plot level, the higher proportion of plots different from their parent transects reflects the pre-eminence of site-specific and tree-specific decisions. As we go down in scale, the percentage of management units with distributions significantly different from the forest-wide distribution increases. These data suggest that the Stoddard–Neel System focuses on managing plantations and blocks, not stands or smaller units, as uneven aged entities. The two-storeyed nature of the diameter distributions at all scales suggest that a forester could manage the smaller units as two-aged (really two-sized) stands.

Discussion and conclusion

Important components of the management system creating these forests include: (1) light partial cuts that keep overall density well below full stocking; (2) patch regeneration and gradual release of the juvenile pines; (3) variation of the age classes, dominance classes, standing stumps and treeless areas within a landscape mosaic; (4)

minimization of mechanical disturbance to native ground cover and soil; and (5) maintenance of the open park-like conditions with frequent prescribed fire (Mitchell *et al.*, 2000).

How did the forest get that way?

Neel states that he used single-tree selection to create and maintain the Stoddard–Neel structure. Standard considerations of negative (improvement) thinning – where you focus on removing poorer quality trees to increase the quality of the residual stand (Burschel and Huss, 1997), releasing established juvenile pines, and improving spacing – were used to aid the forester in making his decision (Mitchell *et al.*, 2000). Neel states that the forest grows 4 per cent per year (Leon Neel, personal communication) and that ~90 per cent of annual growth can be harvested in periodic removals (Neel, 1993). What is unique about the system, however, is the emphasis on retaining large, old trees. As there is only a fixed amount of growing space, other parts of the age (size) class distributions must occupy a proportionately smaller area. If the forester emphasizes retaining large trees, and preferentially removes trees that either exhibit poor vigour or poor quality, the marking begins to resemble a thinning from below for the overstorey.

As the plantations were allowed to accumulate volume and new age classes of trees during the first half of the twentieth century, we can assume that the diameter (and perhaps age) distribution was normal at the inception of the Stoddard–Neel system in the 1940s. One could create and then maintain the Stoddard–Neel structure with a thinning (thinning the overstorey from below). We observed that longleaf pine crowns were not very dense, so the combination of large and frequent sunflecks through the canopy and the openings created by timber harvest or quail management would release growing space to the understorey. A low thinning with quality and density/spacing considerations is not as difficult to teach as a true multi-aged reverse-J marking and would allow foresters to quickly begin to convert even-aged stands. The value of the Stoddard–Neel system is that it maintains a continuous forest canopy while maintaining an open midstorey, critical for northern bobwhite quail, red-cockaded woodpecker and understorey plant

diversity. A forest maintained under the BDq system with a constant ratio between successive diameter classes would require a very flat distribution (a very low 'q') to satisfy these requirements, and would run the risk of insufficient ingrowth at the upper diameter levels due to mortality losses by the smaller trees, particularly under an intensive prescribed burning regime. Longleaf pine is a long-lived species, allowing the forest manager to maintain a vigorous overstorey while 'filtering' trees through the midstorey at a density and rate compatible with maintaining the hunting and ecological attributes of the Red Hills plantations.

The key points of the system include maintaining densities below 15 m² ha⁻¹, managing the overstorey with removals from below, maintaining a reproduction component in the stand and allowing transition from reproduction to overstorey on some small proportion of the area.

Acknowledgements

The authors wish to acknowledge the support of the Turner Foundation, Tall Timbers Research Station and the Red Hills plantation community. We are also grateful for the many summer interns' and visiting scholars' contributions to the collection of the data.

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